# Evolving Fusion-Based Visibility Restoration Model for Hazy Remote Sensing Images Using Dynamic Differential Evolution

Dilbag Singh, *Senior Member, IEEE*, Manjit Kaur, *Member, IEEE*, Mohamed Yaseen Jabarulla, *Member, IEEE*, Vijay Kumar, and Heung-No Lee, *Senior Member, IEEE*

***Abstract*— Remote sensing images taken during poor environmental conditions are degraded by the scattering of atmospheric particles, which affects the performance of many imaging systems. Hence, an efficient visibility restoration model is required to remove haze from distorted images. However, the design of visibility restoration models is an ill-posed problem as the physical information, such as depth information and attenuation model, is usually unknown. The physical parameters computed using existing models, such as dark channel prior and gradient channel prior, are not accurate, especially for images with large haze gradients. Therefore, in this article, an evolving visibility restoration model is proposed for remote sensing images. Initially, the fusion-based transmission map is computed from the foreground and sky regions. The transmission map is further improved by designing a hybrid constraint-based variational model. Finally, a dynamic differential evolution is utilized to optimize the control parameters of the proposed model. The proposed model is validated on 50 synthetic benchmarks and 50 real-life remote sensing images. For comparative analysis, ten well-known restoration models are also considered. The comparative analysis demonstrates that the proposed model outperforms the existing restoration models.**

***Index Terms*— Attenuation model, dark channel prior (DCP), depth information, dynamic differential evolution, evolving model, fusion-based transmission map, gradient channel prior, haze, large haze gradient, remote sensing images, variational model, visibility restoration.**

## I. Introduction

In RECENT years, remote sensing images are extensively utilized in various applications. Since optical remote sensors are far from the surface of the Earth through the atmosphere layer, therefore, these images suffer from poor spatial and spectral information issues [1], [2]. Even images captured during the sunny environment also contain some amount of haze due to the higher distance between satellite camera and Earth [3], [4]. The haze, fog, or smog attenuates the actual object's radiance as the distance between the Earth surface and satellite camera increases [5], [6]. An optical imaging model can be represented as [7], [8]

$$S_i(i, j) = A_r(i, j)t(i, j) + (1-t(i, j))a_v \tag{1}$$

where $S_i$ represents a hazy image, $A_r$ shows the actual scene radiance of the image, and $t$ and $a_v$ denote transmission map and global atmospheric veil, respectively. The primary objective of the single-image haze restoration model is to estimate $A_r$ from $S_i$. However, $t$ and $a_v$ are unknown [9], [10]. Therefore, the efficient estimation of $t$ and $a_v$ is required to obtain the haze-free images [5]. The coordinates of pixel values in the image are represented by using $(i, j)$.

Multiple-image-based dehazing models can efficiently restore the hazy images [11]. These models need physical features of actual images in a prior [12]–[14]. However, in real-time applications, multiple images and physical features of the same scene are not available [15], [16]. Therefore, researchers have designed and implemented various single-image dehazing models. The depth information was utilized to estimate the physical feature of hazy images [17], [18]. However, depth information models suffer from halo artifacts, gradient reversal artifacts, texture distortion, edge, and pixel degradation. Dark channel prior (DCP) was proposed to overcome these problems [7]. DCP assumes that the intensity value of at least one color channel is near to zero in clear days. However, DCP was effective only for images that have a low density of haze [19]. Therefore, depth map-based restoration models, such as bounded DCP [20], sparse DCP [21], adaptive bichannel priors [22], boosting dark channel [23], gradient channel prior [24], and DCP based variational model [25], were proposed. However, these models suffer from gradient and texture distortion problems.

To obtain more efficient results, a fusion-based defogging model (FDM) was implemented [26]. Gaussian-based DCP was used to estimate atmospheric light. To prevent overenhancement in the sky region, contrast limited adaptive histogram equalization (CLAHE) was used to improve the fine

Manuscript received November 11, 2021; revised February 17, 2022; accepted February 26, 2022. Date of publication March 2, 2022; date of current version April 5, 2022. This work was supported in part by the National Research Foundation of Korea (NRF) Grant funded by the Korean Government (MSIP) under Grant NRF-2021R1A2B5B03002118; and in part by the Ministry of Science and ICT (MSIT), Korea, through the Information Technology Research Center (ITRC) Support Program supervised by the Institute of Information and Communications Technology Planning and Evaluation (IITP) under Grant IITP-2021-0-01835. *(Corresponding author: Heung-No Lee.)*

Dilbag Singh, Manjit Kaur, Mohamed Yaseen Jabarulla, and Heung-No Lee are with the School of Electrical Engineering and Computer Science, Gwangju Institute of Science and Technology, Gwangju 61005, South Korea (e-mail: dggill2@gmail.com; manjit.kr@yahoo.com; yaseen@gm.gist.ac.kr; heungno@gist.ac.kr).

Vijay Kumar is with the Computer Science and Engineering Department, National Institute of Technology, Hamirpur, Himachal Pradesh 177005, India (e-mail: vijaykumarchahar@gmail.com).

Digital Object Identifier 10.1109/TGRS.2022.3155765

1558-0644 © 2022 IEEE. Personal use is permitted, but republication/redistribution requires IEEE permission.
See https://www.ieee.org/publications/rights/index.html for more information.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 22:56:45 UTC from IEEE Xplore. Restrictions apply.

details by keeping the color fidelity. A flicker-free module was also used to minimize the flickering effect that occurred due to frame-based dehazing. Although FDM achieves better results, it suffers from poor computational speed and texture distortion. Variational minimization-based dehazing techniques were also designed to obtain the transmission map and atmospheric veil [27]. However, these models are not effective against the dense haze. Therefore, an evolving restoration model is proposed to restore the hazy images.

The novelty of this research work is summarized as follows.

1) An evolving restoration model is proposed for restoring the haze from remote sensing images.
2) A fusion-based transmission map is proposed by integrating the transmission map obtained from both luminance-based channel prior and oblique gradient-based profile prior.
3) A hybrid constraints-based variational minimization is proposed to improve the computed transmission map.
4) Finally, dynamic differential evolution is used to tune the initial parameters of the proposed model.
5) The proposed haze removal model can successfully overcome sky-region and texture distortion issues with an existing haze restoration model.

The remaining article is structured as follows. The related work is presented in Section II. The mathematical detail of the proposed model is mentioned in Section III. The evolving proposed visibility restoration model is presented in Section IV. Experimental results and discussion are mentioned in Section V. Section VI concludes the remarks.

## II. Related Works

This section presents a review of some well-known image dehazing models. The objective is to evaluate various features of the existing models.

Ancuti and Ancuti [28] fused the results of contrast enhancement and white balance for restoring the images. Three weighted maps, i.e., saliency, chromaticity, and luminance, were also used to preserve the potential regions. To reduce artifacts, a Laplacian pyramid-based multiscale fashion was also used. Although this method has shown better results, it suffers from the texture distortion issue. Wang *et al.* [29] proposed region decomposition and feature fusion for restoring the images with a large sky region. Gradient and grayscale detail-based quad-tree was used to compute the sky subregion. A weight map was used to refine the transmission map. Color compensation and atmospheric scattering model were used to obtain the restored images. However, this model suffers from the gradient reversal artifacts problem.

Wang *et al.* [30] designed a fast linear transformation (FLM)-based dehazing model. Quad-tree subdivision was used to estimate the atmospheric light. An atmospheric scattering model was then used to restore the images. FLM has shown remarkable computational speed. Zhang *et al.* [31] designed a pyramid channel-based feature attention network (PFAN) to remove the haze from the image. PFAN was used to preserve the features at different levels. To achieve this, the three-scale feature extraction module was used. To remove the haze, PFAN was used to obtain the channel maps and combine the vital features in a pyramid manner. Zheng *et al.* [32] applied an adaptive structure decomposition integrated multiexposure image fusion (AMEF) to restore the images. Gamma correction was used to extract the underexposed image sequences. The multiexposure image fusion model was applied to the different exposure-level images to obtain the haze-free image. However, FLM, PFAN, and AMEF suffer from the texture distortion problem.

Li *et al.* [33] implemented the image dehazing technique using a convolutional neural network (CNN). The latent dehazed image was reconstructed by using a perception-inspired haze removal subnetwork. Joint multiterm loss optimization was applied to improve the properties of the dehazed image through a refinement network. Zhang *et al.* [34] studied an image dehazing based on a deep learning model (DLM). Joint estimation was utilized to estimate the transmission map. The halos and artifacts were eliminated by using global regularization. However, CNN- and DLM-based dehazing models suffer from poor computational speed.

Ju *et al.* [35] proposed an image dehazing technique based on gamma correction prior (IDGCP). The globalwise scheme was used to implement IDGCP. It has reduced the computation time because it needs to find only one unknown constant to obtain good quality restoration without applying the refining process. Zhao *et al.* [36] proposed an image dehazing technique using a multiscale optimal fusion model (MOFM). The transmission region was estimated by fusing maskwise and pixelwise transmission maps. Halo artifacts were suppressed by embedding MOF into maskwise dehazing. Kim *et al.* [37] utilized saturation-based transmission map (STM) estimation to restore the images. A color veil removing algorithm was used for the images that have fine or yellow dust. Huang *et al.* [38] proposed a self-supporting dehazing network (SSDN) to restore the images. A self-filtering block was used to eliminate the superfluous features. A self-supporting module was applied to recover the details of an image. However, IDGCP, MOFM, STM, and SSDN suffer from the texture distortion problem.

Golts *et al.* [39] implemented an unsupervised DCP (uDCP) as an energy function to restore the images. The parameters of network were tuned by minimizing the uDCP. Ancuti *et al.* [40] improved the performance of image dehazing techniques using color channel transfer (CCT). The information loss was compensated by applying a color-transfer strategy to balance the chromatic loss. Yin *et al.* [41] solved the issues associated with physical parameter estimation of imaging. The channel and spatial features were captured using an attention block-based dehazing network. The essential features of spatial positions were captured using the spatial attention module. The channelwise features' interdependencies were exploited using the channelwise module. Wang *et al.* [42] restored the images by applying the haze concentration adaptive network (HCAN). A pyramid feature extractor was utilized to influence the features of CNN layers to improve an image. Feature enhancement and multiscale feature attention modules were also used to obtain a clear

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 22:56:45 UTC from IEEE Xplore. Restrictions apply.

TABLE I
COMPARATIVE ANALYSIS OF THE COMPETITIVE VISIBILITY RESTORATION MODELS

| Ref. No. | Year | model | Edge preservation | Texture preservation | Color distortion | Halo artifacts | Large haze gradients | Gradient reversal artifacts |
|---|---|---|---|---|---|---|---|---|
| [7] | 2011 | DCP | ✗ | ✗ | ✗ | ✓ | ✗ | ✗ |
| [43] | 2012 | Wiener filter | ✗ | ✗ | ✓ | ✗ | ✗ | ✗ |
| [44] | 2013 | DCP and contrast | ✗ | ✗ | ✗ | ✓ | ✓ | ✓ |
| [45] | 2014 | Segment DCP | ✗ | ✗ | ✗ | ✓ | ✓ | ✗ |
| [46] | 2015 | DCP | ✗ | ✗ | ✓ | ✗ | ✓ | ✗ |
| [27] | 2016 | Adaptive variational | ✗ | ✗ | ✓ | ✗ | ✗ | ✗ |
| [30] | 2016 | Fast linear transform | ✓ | ✗ | ✗ | ✗ | ✗ | ✗ |
| [26] | 2017 | FDM | ✗ | ✗ | ✓ | ✗ | ✗ | ✗ |
| [22] | 2017 | Adaptive bi-channel priors | ✗ | ✗ | ✗ | ✓ | ✓ | ✗ |
| [33] | 2019 | CNN | ✗ | ✗ | ✗ | ✓ | ✓ | ✗ |
| [39] | 2019 | uDCP | ✗ | ✗ | ✗ | ✓ | ✓ | ✓ |
| [20] | 2019 | Bounded optimization | ✗ | ✗ | ✓ | ✗ | ✗ | ✗ |
| [47] | 2019 | PMS-Net | ✓ | ✓ | ✗ | ✗ | ✓ | ✗ |
| [32] | 2020 | AMEF | ✗ | ✗ | ✗ | ✓ | ✓ | ✗ |
| [31] | 2020 | PFAN | ✗ | ✗ | ✗ | ✓ | ✓ | ✗ |
| [24] | 2020 | Gradient channel prior | ✓ | ✓ | ✗ | ✗ | ✓ | ✗ |
| [21] | 2020 | Sparse DCP | ✗ | ✗ | ✗ | ✗ | ✓ | ✓ |
| [23] | 2020 | Boosting DCP | ✓ | ✗ | ✗ | ✗ | ✗ | ✓ |
| [48] | 2020 | DFM | ✓ | ✗ | ✗ | ✗ | ✓ | ✓ |
| [49] | 2020 | PHLD | ✓ | ✗ | ✗ | ✗ | ✓ | ✓ |
| [25] | 2020 | Guided variational model | ✗ | ✗ | ✗ | ✓ | ✓ | ✓ |
| [50] | 2021 | IASM | ✓ | ✗ | ✗ | ✓ | ✓ | ✗ |

image with better details. However, uDCP, CCT, and HCAN suffer from texture distortion and poor computational speed issues.

Chen *et al.* [47] implemented an efficient patch map selection network (PMS-Net) to predict the patch size for every pixel. CNN was used to obtain a patch map from image to image. A multiscale U-module with a pyramid style was also designed to efficiently predict the transmission map and atmospheric light. Chen *et al.* [49] designed a patch-map-based hybrid learning DehazeNet (PHLD) model that has combined biattentive generative adversarial network with patch map. The patch map was used to adaptively select patch size. A patch map-based DCP model was also embedded in PHLD.

Ju *et al.* [50] applied an improved atmospheric scattering model (IASM) to restore the images. The dim effect was resolved by using the light absorption coefficient in IASM. Gao *et al.* [51] implemented an image dehazing model using the self-constructing image fusion method (SCFM). Segmentation technique and $L_0$ gradient optimization were used to obtain atmospheric light and transmission map. Guo *et al.* [52] restored the images using CNN with the fusion method. A neural network was used to obtain a haze-free image. Gao *et al.* [48] designed a dual-fusion model (DFM) to obtain better dehazing results. The segmentation process was utilized to divide the regions, such as sky and nonsky. A multiregion fusion model was utilized to efficiently compute the transmission map. However, IASM, SCFM, and DFM suffer texture and gradient distortion issues.

As known in prior, the objective of dehazing is to preserve edges, color, and texture information. In addition, the dehazing technique should be able to restore images with a large haze gradient. Also, the computed restored images should not suffer from gradient reversal and halo artifacts issues. Table I demonstrates the comparison among the existing dehazing models based on some potential features of restoration models. The aforementioned restoration model provides significant results but is not so effective against images with a high density of haze. The existing models suffer from various artifacts, such as gradient reversal and halo artifacts. In addition, the tuning of control parameters was achieved through the trial and error method. ✗ and ✓symbols indicate whether the given parameter is considered by the respective dehazing technique or not.

## III. PROPOSED MODEL

This section presents the proposed visibility restoration model. Fig. 1 shows a step-by-step process of the proposed haze removal model. Initially, a single hazy image is used as an input for the proposed model. The main objective of this model is to obtain the restored image. Oblique gradient channel prior is used to estimate the depth information of the hazy image. Thereafter, the physical attributes, such as transmission map and atmospheric veil, are estimated by using the computed oblique gradient map. In addition, the transmission map is also computed from the luminance-based approach. The fused transmission map is then computed by considering both transmission maps obtained from the oblique gradient channel prior and the luminance map. A novel hybrid constraint-based variational model is then utilized to refine the fused transmission map. Finally, the restored image can be computed by utilizing the modified visibility restoration model.

### A. Transmission Map Estimation

Images obtained during sunny weather have foreground and background (i.e., sky) regions. The transmission maps

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 22:56:45 UTC from IEEE Xplore. Restrictions apply.

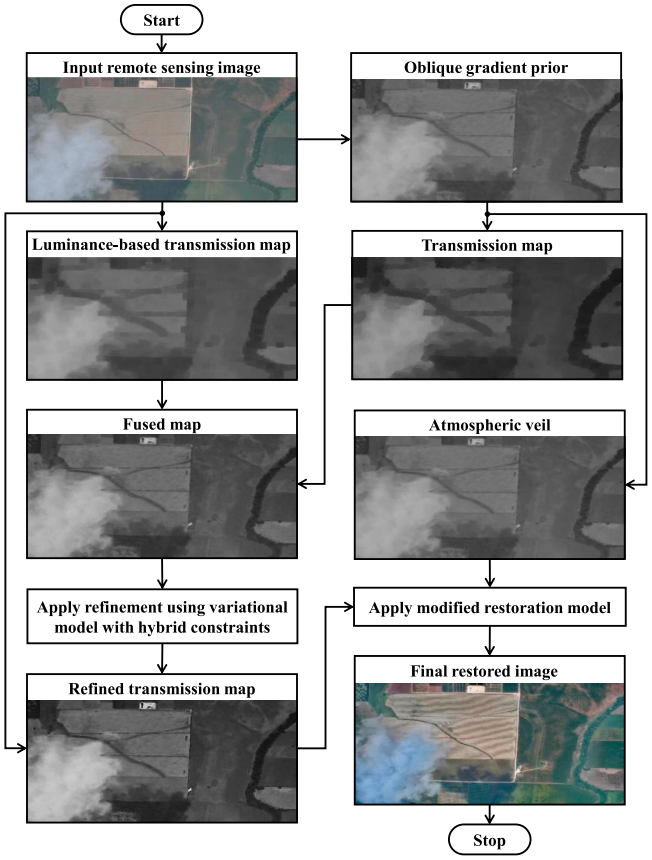


Fig. 1. Proposed haze restoration model for remote sensing images.

computed from DCP [7] and gradient profile prior [53] are not so effective when there exists a large haze gradient in hazy images. Therefore, an oblique gradient profile prior (OGPP) can be used to handle this issue.

*1) Oblique Gradient Profile Prior:* The detailed description of OGPP is given as follows.

An OGPP [54] utilizes the direction and magnitude details of a hazy image ($S_i$). The mathematical formulation is given as follows:

$$\vec{S}_i(j) = \begin{pmatrix} \Upsilon_m \\ \Upsilon_n \end{pmatrix} = \begin{pmatrix} \dfrac{\partial S_i}{\partial m} \\ \dfrac{\partial S_i}{\partial n} \end{pmatrix} \tag{2}$$

where $m$ and $n$ show the $x$- and $y$-axes of an image, respectively. For $S_i(m,n)$, $\Upsilon_m$ and $\Upsilon_n$ can be estimated by considering various masks. An amplitude of $\vec{S}_i$ can be computed as

$$\text{mag}(\vec{S}_i) = \sqrt{(\Upsilon_m^2 + \Upsilon_n^2)}. \tag{3}$$

The orientation angle ($\nu$) of $\vec{S}_i$ can be evaluated as

$$\nu_{\vec{S}_i}(j) = \arctan\left(\frac{\Upsilon_n}{\Upsilon_m}\right). \tag{4}$$

A mask ($\Omega$) having size of $3 \times 3$, which is shown in Fig. 2(a). The central pixel's intensity is defined by $S_{ic}$. $S_{ik}(k = 1, 2, \ldots, 8)$ shows the $k$th sibling of $S_{ic}$. OGPP can approximate nine oblique edges from $S_i$ with $3 \times 3$ mask, whereas the traditional gradient operator fails to extract significant oblique edges [55].

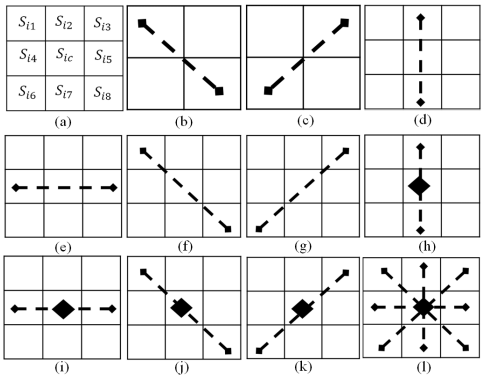


Fig. 2. Oblique gradient-based mask and 11 oblique edges with different mask sizes. (a) $3 \times 3$ mask. (b) Primary diagonal oblique edge with $2 \times 2$ mask. (c) Secondary diagonal oblique edge with $2 \times 2$ mask. (d) Vertical orthogonal edge for $3 \times 3$ mask without center. (e) Horizontal orthogonal edge for $3 \times 3$ mask without center. (f) Primary diagonal oblique edge with $3 \times 3$ mask without center. (g) Secondary diagonal oblique edge with $3 \times 3$ mask without center. (h) Vertical orthogonal edge for $3 \times 3$ mask with center. (i) Horizontal orthogonal edge for $3 \times 3$ mask with center. (j) Primary diagonal oblique edge with $3 \times 3$ mask and center. (k) Secondary diagonal oblique edge with $3 \times 3$ mask and center. (l) Oblique gradient channel prior with center.

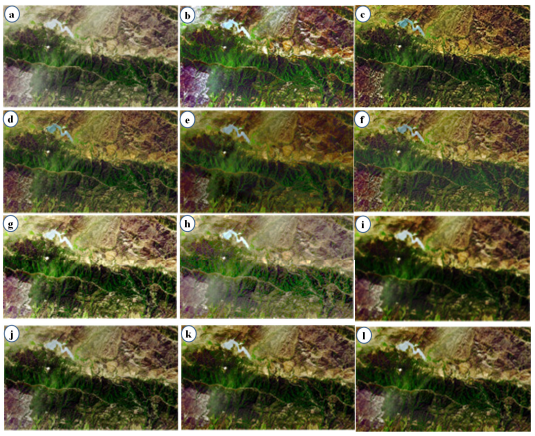


Fig. 3. Visual analysis of 11 oblique edges with different mask sizes on the image along with new visible edges ($e$) values: (a) hazy image, (b) obtained from Fig. 2(b) and $e = 1.96$, (c) obtained from Fig. 2(c) and $e = 1.89$, (d) obtained from Fig. 2(d) and $e = 2.48$, (e) obtained from Fig. 2(e) and $e = 2.39$, (f) obtained from Fig. 2(f) and $e = 2.74$, (g) obtained from Fig. 2(g) and $e = 2.37$, (h) obtained from Fig. 2(h) and $e = 2.71$, (i) obtained from Fig. 2(i) and $e = 2.62$, (j) obtained from Fig. 2(j) and $e = 2.68$, (k) obtained from Fig. 2(k) and $e = 2.78$, and (l) obtained from Fig. 2(l) and $e = 3.24$.

Figs. 3 and 4 show the visual analysis and zoomed results obtained from the oblique edges along with respective new visible edges. It is found that the fused oblique edges [see Fig. 4(k)] achieve remarkably good visibility. Also, it achieves 2.7362% improvements over the other oblique gradient edges.

Fig. 2 shows 11 oblique edges. Fig. 2(b) and (c) shows the oblique edges with $2 \times 2$ mask. Fig. 2(b) and (c) shows possible oblique edges as $(\pi/4) \leftrightarrow (5\pi/4)$ and $(3\pi/4) \leftrightarrow (7\pi/4)$, respectively. By considering $\Omega \in \mathbb{R}^{3\times3}$, four $2 \times 2$ masks are computed as $\begin{bmatrix} S_{i1} & S_{i2} \\ S_{i4} & S_{ic} \end{bmatrix}$, $\begin{bmatrix} S_{i2} & S_{i3} \\ S_{ic} & S_{i5} \end{bmatrix}$, $\begin{bmatrix} S_{i4} & S_{ic} \\ S_{i6} & S_{i7} \end{bmatrix}$, and $\begin{bmatrix} S_{ic} & S_{i5} \\ S_{i7} & S_{i8} \end{bmatrix}$ with respect to $S_{ic}$. The respective OGPP angles can be computed by considering (5)–(8). If $\Delta^1(m,n) = 0$, then the oblique edge [i.e., $(\pi/4) \leftrightarrow (5\pi/4)$] exists in mask $\begin{bmatrix} S_{i1} & S_{i2} \\ S_{i4} & S_{ic} \end{bmatrix}$.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 22:56:45 UTC from IEEE Xplore. Restrictions apply.

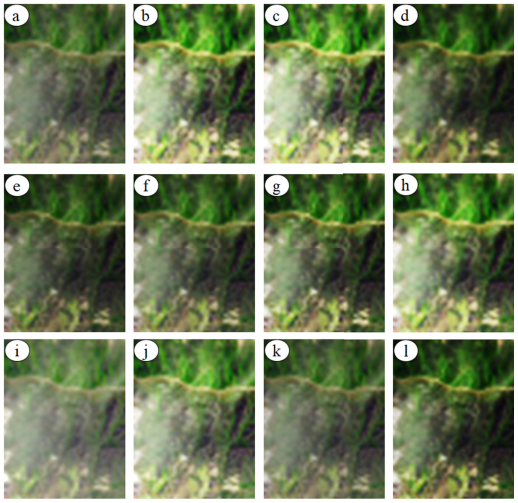


Fig. 4. Zoomed visual analysis of 11 oblique edges with different mask sizes on the image along with new visible edges ($e$) values: (a) hazy image, (b) obtained from Fig. 2(b) and $e = 1.96$, (c) obtained from Fig. 2(c) and $e = 1.89$, (d) obtained from Fig. 2(d) and $e = 2.48$, (e) obtained from Fig. 2(e) and $e = 2.39$, (f) obtained from Fig. 2(f) and $e = 2.74$, (g) obtained from Fig. 2(g) and $e = 2.37$, (h) obtained from Fig. 2(h) and $e = 2.71$, (i) obtained from Fig. 2(i) and $e = 2.62$, (j) obtained from Fig. 2(j) and $e = 2.68$, (k) obtained from Fig. 2(k) and $e = 2.78$, and (l) obtained from Fig. 2(l) and $e = 3.24$.

If $\Delta^1(m,n)$ models to $(\pi/2)$ (or $-(\pi/2)$), then the corresponding edge is evaluated [along $(3\pi/4) \leftrightarrow (7\pi/4)$] in mask $\begin{bmatrix} S_{i1} & S_{i2} \\ S_{i4} & S_{ic} \end{bmatrix}$. Similarly, other oblique edges can be computed from other masks. Fig. 2(d)–(k) represents eight possible oblique gradients using $3 \times 3$ mask size. Fig. 2(l) shows a fused OGPP operator. In Fig. 2(d)–(k), remaining oblique edges [i.e., $(\pi/4) \leftrightarrow (5\pi/4)$, $(3\pi/4) \leftrightarrow (7\pi/4)$, $0 \leftrightarrow \pi$ and $(\pi/2) \leftrightarrow (3\pi/2)$] are computed. OGPPs can be approximated by utilizing (9) and (12). $S_{ic}$ is utilized by using (9) and (10), but not from (11) and (12)

$$\Delta^1(m,n) = \arctan\left(\frac{\Upsilon_n^1}{\Upsilon_m^1}\right) = \arctan\left(\frac{S_{i1} - S_{ic}}{S_{i2} - S_{i4}}\right) \tag{5}$$

$$\Delta^2(m,n) = \arctan\left(\frac{\Upsilon_n^2}{\Upsilon_m^2}\right) = \arctan\left(\frac{S_{i3} - S_{ic}}{S_{i2} - S_{i5}}\right) \tag{6}$$

$$\Delta^3(m,n) = \arctan\left(\frac{\Upsilon_n^3}{\Upsilon_m^3}\right) = \arctan\left(\frac{S_{i6} - S_{ic}}{S_{i7} - S_{i4}}\right) \tag{7}$$

$$\Delta^4(m,n) = \arctan\left(\frac{\Upsilon_n^4}{\Upsilon_m^4}\right) = \arctan\left(\frac{S_{i8} - S_{ic}}{S_{i7} - S_{i5}}\right) \tag{8}$$

$$\Delta^5(m,n) = \arctan\left(\frac{\Upsilon_n^5}{\Upsilon_m^5}\right) = \arctan\left(\frac{S_{i2} + S_{i7} - 2 \times S_{ic}}{S_{i4} + S_{i5} - 2 \times S_{ic}}\right) \tag{9}$$

$$\Delta^6(m,n) = \arctan\left(\frac{\Upsilon_n^6}{\Upsilon_m^6}\right) = \arctan\left(\frac{S_{i1} + S_{i8} - 2 \times S_{ic}}{S_{i3} + S_{i6} - 2 \times S_{ic}}\right) \tag{10}$$

$$\Delta^7(m,n) = \arctan\left(\frac{\Upsilon_n^7}{\Upsilon_m^7}\right) = \arctan\left(\frac{S_{i2} - S_{i7}}{S_{i4} - S_{i5}}\right) \tag{11}$$

$$\Delta^8(m,n) = \arctan\left(\frac{\Upsilon_n^8}{\Upsilon_m^8}\right) = \arctan\left(\frac{S_{i1} - S_{i8}}{S_{i3} - S_{i6}}\right). \tag{12}$$

Fig. 2(l) shows all siblings of $S_{ic}$ and estimates a fused gradient operator. Thus, OGPP is defined as

$$\Delta^9(m,n) = \arctan\left(\frac{\Upsilon_n^9}{\Upsilon_m^9}\right) = \arctan\left(\frac{\sum_{k=1}^{8}(S_{ic} - S_{ik})}{8}\right). \tag{13}$$

An OGPPs ($\Delta$) of a image is evaluated by using (5)–(13). arctan(.) is considered to approximate $\Delta^s(m,n)$. The pixel angle's range in orientation image ($\theta(m,n)$) is assumed to be $((-\pi/2),(\pi/2))$.

The OGPP-based transmission map ($\bar{t}_d$) can be computed as

$$\bar{t}_{\mathrm{d}}(i,j) = 1 - \omega\Delta^9_{c\in\{r,g,b\}}\left(\Delta^9_{\mathrm{y}\in\Omega(i,j)}\left(\frac{S_{ic}(i,j)}{a_{vc}}\right)\right). \tag{14}$$

Here, mask $\Omega(i,j)$ is centered at $i,j$ and $\omega = 0.95$ is a control attribute.

*2) Luminance-Based Transmission Map:* A luminance-based transmission map ($\bar{t}_l$) can be evaluated as [19]

$$\bar{t}_l(i,j) = \exp(-\Delta\hat{L}(i,j)). \tag{15}$$

Here, $\hat{L}$ indicates a modified luminance value and $\Delta$ represents a scattering coefficient. For red, green, and blue channels, $\Delta$ is set to be 0.3, 0.3, and 0.4, respectively (please refer to [19]).

To estimate the impact of depth information on transmission map, a modified luminance ($\hat{L}$) can be rewritten as

$$\hat{L}(i,j) = \frac{\tau}{L^*}L(i,j). \tag{16}$$

Here, $\tau$ shows the range of depth map. $L^*$ determines the 95% percentile value of the luminance ($L$) of $S_i$. Weighted fusing of $\bar{t}_d$ and $\bar{t}_l$ can be used to approximate coarse transmission map ($\bar{t}$) and it can be defined as

$$\bar{t}(i,j) = \chi(i,j)\bar{t}_d(i,j) + (1 - \chi(i,j))\bar{t}_l(i,j) \tag{17}$$

where $\chi \in [0,1]$ shows transmission weight. If pixel $i,j \in \Omega$ belongs to foreground objects, then $\chi(i,j)$ approaches toward 1 and $\bar{t}(i,j) \to \bar{t}_d(i,j)$. Similarly, $\chi(i,j)$ approaches toward 0 and $\bar{t}(i,j) \to \bar{t}_l(i,j)$.

The weight function ($\chi$) is defined as

$$\chi(i,j) = \frac{1}{1 + e^{-\theta_1\bar{t}_d(i,j) - \theta_2}}. \tag{18}$$

Also,

$$\theta_1 = \frac{l_1}{\max(\bar{t}_d) - \min(\bar{t}_d)} \tag{19}$$

and

$$\theta_2 = -l_2 - \theta_1 \times \min(\bar{t}_d). \tag{20}$$

$l_1$ and $l_2$ are the control parameters of the weight function for luminance-based transmission map. However, $\bar{t}$ may contain some noise, and also, it may not be so accurate. Therefore, a variational model is designed to improve $\bar{t}$.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 22:56:45 UTC from IEEE Xplore. Restrictions apply.

### B. Variational Model

The imaging model from (1) can be rewritten as [56]

$$\bar{S}_i(i,j) = \bar{A}_r(i,j)t(i,j). \tag{21}$$

Here,

$$\bar{S}_i(i,j) = a_v - S_i(i,j) \tag{22}$$

and

$$\bar{A}_r(i,j) = a_v - A_r(i,j). \tag{23}$$

Here, $i, j \in \Omega$. Also, $\bar{A}_r$ can be defined as

$$\bar{A}_r(i,j) = \frac{a_v - S_i(i,j)}{\max(\bar{t}(i,j), t_\varepsilon)}. \tag{24}$$

To prevent imaging instability, lower bound ($t_\varepsilon$) is used. For deducing (24), $S_i = S_{ic}$, $\bar{S}_i = \bar{S}_{ic}$, and $\bar{A}_r = \bar{A}_{rc}$ for $c \in \{\text{Red}(r), \text{Green}(g), \text{Blue}(b)\}$.

To refine the transmission map, a novel hybrid regularization-based variational model is defined as

$$\min_{\bar{A}_r,t} \left\{ \frac{\psi_1}{2}\left\|\bar{S}_i - \bar{A}_r t\right\|_2^2 + \frac{\psi_2}{2}\|t-\bar{t}\|_2^2 + \psi_3\left\|P \circ \left(\nu t - \vec{S}_i\right)\right\|_1 + \psi_4\left\|\nu\bar{A}_r\right\|_1 + \psi_5\|\nu t\|_1 \right\}. \tag{25}$$

Here, regularization is achieved by using $\psi_{1\le i\le 4}$. $\psi$ defines the variational minimization parameter. In (25), initial terms define a squared $L_2$-norm. Thereafter, the $L_1$-norm is utilized to preserve the edge information. The $L_1$-norm is required as it does not require any parameter to control the degree of flatness as in the case of $L_0$ gradient term to the quadratic data-fidelity term. $\nu$ defines the orientation angle. The last term defines total variation (TV), which may improve the estimation process. A weighting operator ($P$) is defined as

$$P = e^{-\gamma\left\|\vec{S}_i\right\|_2^2}. \tag{26}$$

Here, $\gamma$ is used to distinguish between the potential edges and remaining pixels to minimize various artifacts. To overcome the nonsmooth $L_1$-norm penalties in (25), a switching model is also considered by introducing three operators, such as $X = \nu t - \vec{S}_i$, $Y = \nu\bar{A}_r$, and $Z = \nu t$. Afterward, (25) can be rewritten to confined variation as

$$\min_{X,i,j,\bar{A}_r,t} \left\{ \frac{\psi_1}{2}\left\|\bar{S}_i - \bar{A}_r t\right\|_2^2 + \frac{\psi_2}{2}\|t-\bar{t}\|_2^2 + \psi_3\|P\circ X\|_1 + \psi_4\|Y\|_1 + \psi_5\|Z\|_1 \right\}. \tag{27}$$

The Lagrangian operator is redefined as

$$\mathcal{L}_{\mathcal{A}} = \frac{\psi_1}{2}\left\|\bar{S}_i - \bar{A}_r t\right\|_2^2 + \frac{\psi_2}{2}\left\|t-\bar{t}\right\|_2^2 + \psi_3\left\|P\circ X\right\|_1 + \psi_4\left\|Y\right\|_1 + \psi_5\left\|Z\right\|_1 + \frac{\Delta_1}{2}\left\|X - \left(\nu t - \vec{S}_i\right) - \frac{\kappa}{\Delta_1}\right\|_2^2 + \frac{\Delta_2}{2}\left\|Y - \nu\bar{A}_r - \frac{\alpha}{\Delta_2}\right\|_2^2 + \frac{\Delta_3}{2}\left\|Z - \nu t - \frac{\delta}{\Delta_3}\right\|_2^2. \tag{28}$$

Here, $\kappa$, $\alpha$, and $\delta$ define Lagrangian multipliers and $\Delta_1$, $\Delta_2$, and $\Delta_3$ are nonnegative attributes. An alternating direction can be utilized to decompose $\mathcal{L}_{\mathcal{A}}$ into several assumptions by considering $X$, $Y$, $Z$, $\bar{A}_r$, and $t$. To optimize these alternatives, the following subproblems are defined.

*1) ($X$, $Y$, $Z$) Subproblems:* Assume that $\bar{A}_r$, $t$, and ($X$, $Y$, $Z$) are available in prior, and the subproblems by considering $L_1$-regularized least squares are defined as

$$X \leftarrow \min_X \left\{ \psi_3\left\|P\circ X\right\|_1 + \frac{\Delta_1}{2}\left\|X - \left(\nu t - \vec{S}_i\right) - \frac{\kappa}{\Delta_1}\right\|_2^2 \right\} \tag{29}$$

$$Y \leftarrow \min_Y \left\{ \psi_4\left\|Y\right\|_1 + \frac{\Delta_2}{2}\left\|Y - \nu\bar{A}_r - \frac{\alpha}{\Delta_2}\right\|_2^2 \right\} \tag{30}$$

$$Z \leftarrow \min_Z \left\{ \psi_5\left\|Z\right\|_1 + \frac{\Delta_3}{2}\left\|Z - \nu t - \frac{\delta}{\Delta_3}\right\|_2^2 \right\}. \tag{31}$$

A shrinkage operator ($\mathcal{S}$) is also used to refine these subproblems as

$$X \leftarrow \mathcal{S}\left(\nu t - \vec{S}_i + \kappa/\Delta_1, \psi_3 P/\Delta_1\right) \tag{32}$$

$$Y \leftarrow \mathcal{S}\left(\nu\bar{A}_r + \alpha/\Delta_2, \psi_4/\Delta_2\right) \tag{33}$$

$$Z \leftarrow \mathcal{S}(\nu t + \delta/\Delta_3, \psi_5/\Delta_3). \tag{34}$$

Here, shrinkage ($\mathcal{S}$) operator can be defined as

$$\mathcal{S}(a,b) = \max(|a| - b, 0) \circ \mathbf{s}(a). \tag{35}$$

Here, $\mathbf{s}$ defines the signum function (see [57]).

*2) ($\bar{A}_r$, $t$) Subproblems:* Assume that $X$, $Y$, and $Z$ are computed from earlier iterations, and then, the minimization of $\mathcal{L}_{\mathcal{A}}$ with respect to $\bar{A}_r$ and $t$ is identical to optimize the following least-squares problem as:

$$\begin{cases} \bar{A}_r \leftarrow \min_{\bar{A}_r} \left\{ \frac{\psi_1}{2}\left\|\bar{S}_i - \bar{A}_r t\right\|_2^2 + \frac{\Delta_2}{2}\left\|Y - \nu\bar{A}_r - \frac{\alpha}{\Delta_2}\right\|_2^2 \right\} \\ t \leftarrow \min_t \left\{ \frac{\psi_1}{2}\left\|\bar{A}_r t - \bar{S}_i\right\|_2^2 + \frac{\psi_2}{2}\left\|t-\bar{t}\right\|_2^2 + z\left\|\nu t - \Upsilon\right\|_2^2 \right\}. \end{cases} \tag{36}$$

Here,

$$\Upsilon = \frac{\Delta_1\hat{X} + \Delta_3\hat{Z}}{\Delta_1 + \Delta_3} \tag{37}$$

with

$$\hat{X} = X + \vec{S}_i - \frac{\kappa}{\Delta_1} \tag{38}$$

and

$$\hat{Z} = Z - \frac{\delta}{\Delta_3}. \tag{39}$$

Also,

$$z = \frac{\Delta_1 + \Delta_3}{2}. \tag{40}$$

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 22:56:45 UTC from IEEE Xplore. Restrictions apply.

TABLE II
HYPERPARAMETERS OF THE PROPOSED DEHAZING MODEL

| Parameter | Lower bound | Upper bound | Data type |
|---|---|---|---|
| Oblique edge type | 3 | 11 | Integer |
| Weight function ($w_1$) | 0 | 30 | Integer |
| Weight function ($w_2$) | 0 | 30 | Integer |
| Percentile value ($p_v$) of the luminance ($L$) | 90 | 100 | Integer |
| Lower bound ($t_\varepsilon$) for restoration | 0 | 1 | Float |
| Transmission control ($\omega$) | 0 | 1 | Float |
| Luminance weight function 1 ($l_1$) | -100 | 100 | Integer |
| Luminance weight function 2 ($l_1$) | -100 | 100 | Integer |

Assume that $F_t$ indicates Fast Fourier transform. Thus, $\bar{A}_r$ and $t$ are defined as

$$\bar{A}_r \leftarrow F_t{}^{-1}\left(\frac{\psi_1 F_t(I/t) + \Delta_2\overline{F_t(v)}F_t(Y-\alpha/\Delta_2)}{\psi_1 F_t(I) + \Delta_2\overline{F_t(v)}F_t(v)}\right) \tag{41}$$

$$t \leftarrow F_t{}^{-1}\left(\frac{\psi_1 F_t(I/\bar{A}_r)+\psi_2 F_t(\bar{t})+(\Delta_1+\Delta_3)\overline{F_t(v)}F_t(\Upsilon)}{(\psi_1+\psi_2)F_t(I)+(\Delta_1+\Delta_3)\overline{F_t(v)}F_t(v)}\right). \tag{42}$$

Here, $F_t{}^{-1}(\cdot)$ shows the inverse of fast Fourier transform and $I$ represents the identity matrix. Complex conjugate operator is defined by using $\overline{F_t(\cdot)}$.

*3) $\kappa$, $\alpha$, and $\delta$ Update:* In each evolution, Lagrangian multipliers $\kappa$, $\alpha$, and $\delta$ are defined as

$$\kappa \leftarrow \kappa - \upsilon\Delta_1(X-(\upsilon t - \upsilon I)) \tag{43}$$

$$\alpha \leftarrow \alpha - \upsilon\Delta_2\left(Y - \upsilon\bar{A}_r\right) \tag{44}$$

$$\delta \leftarrow \delta - \upsilon\Delta_3(Z-\upsilon t) \tag{45}$$

where $\upsilon$ defines the length of a step.

### C. Haze Restoration Model

To overcome the textures distortion, a restored image ($A_r$) is obtained by considering $t$ in (42). A haze restoration model can be represented as [54], [58]

$$A_r(i,j) = \frac{S_i(i,j) - a_\upsilon}{\max(t(i,j), t_\varepsilon)} + a_\upsilon \tag{46}$$

where $a_\upsilon$ defines the computed atmospheric veil.

## IV. EVOLVING VISIBILITY RESTORATION MODEL

In this section, the parameter tuning of the proposed visibility restoration model is resolved.

### A. Problem Formulation

The proposed haze removal model has many control parameters. Table II demonstrates various control parameters of the proposed model along with their respective ranges and data type. It clearly shows that the optimization of these parameters is necessary to achieve efficient outcomes. The tuning of these parameters through a trial and error basis may not achievable to produce the desirable results.

### B. Fitness Function

A multiobjective fitness function is used to optimize the control parameters of the proposed model, The performance metrics, i.e., percentage of saturated pixels ($\rho$) and new visible edges ($e$), are utilized to define the multiobjective fitness ($f(t)$) function. It is defined as

$$f(t) = \begin{cases} \text{Min } (\rho) \\ \text{Max } (e). \end{cases} \tag{47}$$

Percentage of saturated pixels ($\rho$) [59] defines the pixels which become either black or white in the restored image, but not white or black in the hazy image. $\rho$ can be formulated as

$$\rho = \frac{S_p}{R \times C} \tag{48}$$

where $S_p$ defines saturated pixels in restore image, which were not black or white in a hazy image. The minimum $\rho$ value shows that the visibility restoration model is better than the others. $R$ and $C$ define the row and columns of a hazy image, respectively.

The ratio of new visible edges ($e$) computes the enhanced rate of visible edges from restored images. $e$ is desirable to be maximum. It can be defined as [60]

$$e = \frac{n_k - n_l}{n_l} \tag{49}$$

where $n_k$ and $n_l$ show the visible edges' cardinal number in hazy image ($S_i$) and the obtained haze-free image ($I_r$), respectively.

To evolve the proposed visibility restoration model, the dynamic differential evolution is utilized as an optimal Pareto group, showing the better trade between the percentage of saturated pixels ($\rho$) and new visible edges ($e$).

### C. Preliminaries

Multiobjective optimization models generally come up with three principle concepts as dominance relation, Pareto optimal group, and Pareto front.

*Definition 1 (Dominance Relation):* Any ($\alpha \in \Theta^\beta$) is said to be dominance vector if other $\gamma \in \Theta^\beta$, depicted as $\alpha > \gamma$, if $\forall n \in \{n = 1, 2, \ldots, \beta\}$, $\alpha_n \leq \gamma_n$, or $\exists\delta \in \{\delta = 1, 2, \ldots, \beta\}$, $\alpha_\delta < \gamma_\delta$.

*Definition 2 (Pareto Optimal Group):* A solution $\eta^* \in \Theta D$ is considered as Pareto optimal to (47), if $\nexists\ \eta$ s.t. $\eta > \eta^*$. A Pareto group ($\Phi^*$) contains all Pareto optimal solutions.

*Definition 3 (Pareto Front):* In multiobjective optimization, a Pareto front ($P_f$) is defined as $P_f = \{f(\eta^*)|\eta^* \in \Phi^*\}$.

### D. Differential Evolution

Storn and Price [61] designed a conventional differential evolution to solve many real-world problems. Crossover, selection, and mutation are the primary operation in differential evolution. It follows the following steps.

*Step 1 (Initialization):* Initially, numerous control parameters, such as population size ($P_s$), mutation rate ($\Psi$), and crossover rate ($c_r$), are defined. Random population is obtained from the feasible region $\Omega$ as $\eta_\delta^0(\delta = 1, 2, \ldots, P_s)$.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 22:56:45 UTC from IEEE Xplore. Restrictions apply.

The evolution step is controlled using a maximum number of iterations ($t_M$).

*Step 2 (Evolution Step):* In this step, to enhance the convergence speed, crossover and mutation operators are utilized.

*Step 2(a) (Mutation):* It is applied on every solution $\eta_\delta^t$ to compute a vector $\gamma_\delta^t$. Some extensively utilized mutation operators are defined as

$$\mathrm{DE}/r_d/1 : \gamma_\delta^t = \eta_r^t + \Psi \cdot \left(\eta_{r2}^t - \eta_{r3}^t\right) \tag{50}$$

$$\mathrm{DE}/b_s/1 : \gamma_\delta^t = \eta_{b_s}^t + \Psi \cdot \left(\eta_r^t - \eta_{r2}^t\right). \tag{51}$$

Here, $r_1$, $r_2$, and $r_3$ are natural numbers randomly selected from $[1, P_s, ]$ and $\neq$ index $\delta$. $\eta_{b_s}^t$ contains the best $b_s$ fitness value during the $t$th iteration.

*Step 2(b) (Crossover):* For each $\eta_\delta^t$, a trial vector $\alpha_\delta^t$ can be obtained as

$$\alpha_{\delta_i}^t = \begin{cases} \gamma_{\delta_i}^t, & \Theta_J \le c_r \text{ or } i = \imath_{r_d} \\ \eta_{\delta_i}^t, & \text{otherwise.} \end{cases} \tag{52}$$

Here, $\Theta_b \in [0,\ 1]$ and $\imath_{r_d} \in [1,\ D]$ are random values.

*Step 3 (Selection):* The trial vector ($\alpha_\delta^t$) is considered with its target vector ($\eta_\delta^t$) as

$$\eta_\delta^{t+1} = \begin{cases} \alpha_\delta^t, & f\left(\alpha_\delta^t\right) \le f\left(\eta_\delta^t\right) \\ \eta_\delta^t, & \text{otherwise.} \end{cases} \tag{53}$$

*Step 4:* Repeat steps 2 and 3 until the termination criteria are not satisfied.

### E. Dynamic Differential Evolution

Tuning of the proposed model's control parameters can be defined as a multiobjective problem. Dynamic differential evolution can optimize these control parameters as follows.

*Step 1 (Initialization):* Initially, define the control attributes of dynamic differential evolution such as $\Psi$, $c_r$, and boundary evaluation ($B_{\text{group}}$) group. Initial solutions can be evaluated as

$$\eta_\delta^o = \eta_{\text{in}} + r_d \cdot (\eta_M - \eta_{\text{in}}) \quad (\delta = 1, 2, \ldots, P_s). \tag{54}$$

Here, $\eta_\delta^o$ represents the initial population, which consists of all solutions. Define generation number to $t = 0$. Define the value of maximum generations ($t_M$). Allocate $t_s = 0.2^* t_M$ (see [62]). A mutation operator pool ($Q^t$) having $P_s/2$ operations is selected from $\mathrm{DE}/r_d/1$ and $\mathrm{DE}/b_s/1$.

*Step 2 (Evolution):* In this step, mutation, boundary evaluation, and crossover operator are applied to obtain new solutions.

*Step 2(a) (Mutation):* For every solution $\eta_\delta^t$, if $t < t_s$, then $\mathrm{DE}/r_d/1$ is used to evaluate solutions; otherwise, a mutation operator is selected randomly from $Q^t$ to compute a vector $\gamma_\delta^t$ and will be recomputed from $Q^t$. Also, the best solution in the mutation model $\mathrm{DE}/b_s/1$ is randomly elected from $P_s$ and computed during the $(t-1)$th iteration.

*Step 2(b) (Boundary Evaluation):* If $r_d > B_{\text{group}}$, then

$$\gamma_{\delta_i}^t = \begin{cases} \eta_{r\imath}^t, & \gamma_{\delta_i} < L_b \\ \eta_{r\imath}^t, & \gamma_{\delta_i} > U_b. \end{cases} \tag{55}$$

Here, $U_b$ and $L_b$ are upper and lower bounds of the $\imath$th attribute of $\eta_\delta$; otherwise, an unfeasible attribute is randomly recomputed from the feasible region $\Omega$.

*Step 2(c) (Crossover):* A trial vector $\alpha_\delta^t$ is computed by using (52).

*Step 3 (Selection):* It can be achieved as follows.

1) If $\eta_\delta^t > \alpha_\delta^t$, then $\eta_\delta^t$ is stored in $P_1^t$.
2) If $\alpha_\delta^t > \eta_\delta^t$, then $\eta_\delta^t$ is replaced by $\alpha_\delta^t$ in $P_1^t$.
3) If $\eta_\delta^t$ and $\alpha_\delta^t$ has no dominated relationship among them, $\alpha_\delta^t$ is then utilized in $P_1^t$.

*Step 4 (Selection):* Crowding distance ($C_d$) sorting [63] is utilized to select solutions. $C_d$ of an individual is evaluated using the mean distance of its two sibling solutions. The minimum and maximum values are set to be infinite, $C_d$, therefore, can be used in every iteration. The final value of $C_d$ is evaluated by concatenating all values of $C_d$. Add $P_s$ to $P_1^t$ depending on $C_d$ values.

*Step 5:* $t = t + 1$

*Step 6:* Repeat steps 2–5 until $t = t_M$.

## V. Performance Analysis

In this section, the performance of the proposed model is evaluated on the well-known benchmark hazy images and compared with the competitive existing techniques.

### A. Experimental Setup

To compute the performance of the proposed haze removal model, 50 synthetic benchmarks and 50 real-life images are obtained from the satellite imaging group [64] and RESIDE [65]. For comparative analysis, ten restoration models are considered. These models are DCP [7], PFAN [31], AMEF [32], CNN [33], uDCP [39], DFM [48], IASM [50], FDM [26], PMS-Net [47], and PHLD [49]. For better comparative analysis, we have also simulated all the models. The parameters of the existing models are selected as they were reported in their respective published articles. The proposed restoration model is tested on a core i7 processor with 16-GB RAM on MATLAB 2019a software.

### B. Visual Analysis of the Proposed Model

Fig. 5 shows the visual analysis between the proposed model with and without the use of dynamic differential evolution on two hazy remote sensing images along with the two performance metrics, i.e., percentage of saturated pixels ($\rho$) and new visible edges ($e$). From Fig. 5(b) and (e), it can be seen that the images obtained from the proposed model without dynamic differential evolution suffer from the oversaturation and texture distortion issues, whereas Fig. 5(c) and (f) clearly shows that the dynamic differential evolution-based proposed model can preserve the texture information with a lesser number of saturated pixels. Also, the obtained quantitative values indicate that the dynamic differential evolution-based proposed model achieves significantly better performance in terms of $\rho$ and $e$.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 22:56:45 UTC from IEEE Xplore. Restrictions apply.

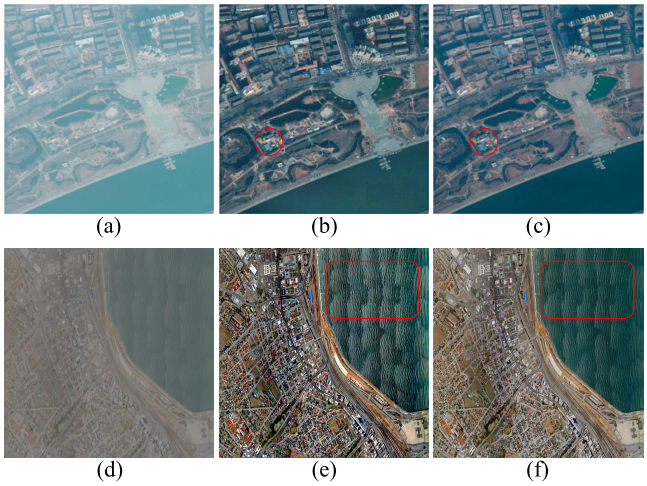


Fig. 5. Visual analysis of the proposed model with and without the use of dynamic differential evolution: (a) input hazy image, (b) without dynamic differential evolution ($\rho = 0.045$ and $e = 2.452$), (c) with dynamic differential evolution ($\rho = 0.012$ and $e = 2.782$), (d) input hazy image, (e) without dynamic differential evolution ($\rho = 0.031$ and $e = 2.017$), and (f) with dynamic differential evolution ($\rho = 0.016$ and $e = 2.548$).

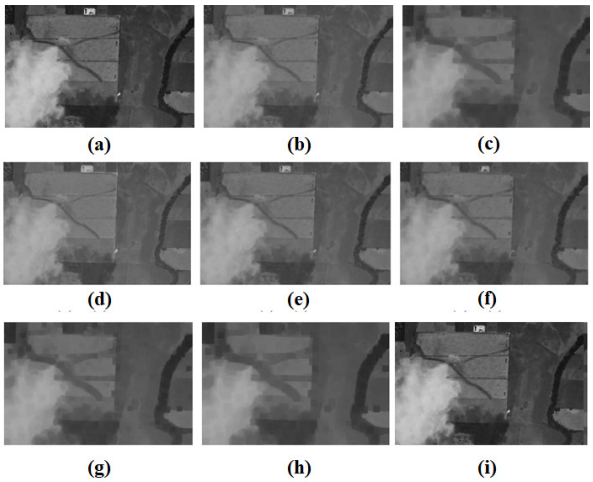


Fig. 6. Comparative analysis of transmission map: (a) ground truth, (b) DCP [7], (c) PFAN [31], (d) AMEF [32], (e) CNN [33], (f) uDCP [39], (g) DFM [48], (h) IASM [50], and (i) Proposed.

### C. Comparative Visual Analysis

Fig. 6(a) shows the actual depth information of hazy image. The improved transmission map computed using DCP [7] is shown in Fig. 6(b). However, it contains some noise, and therefore, transmission map is not smooth. Fig. 6(c) and (g) shows the transmission maps computed from PFAN [31] and DFM [48], respectively. These maps seem oversaturated. Fig. 6(d) and (f) shows the refined transmission map obtained from AMEF [32] and uDCP [39], respectively. However, these maps seem to be over enhanced and, thus, may suffer from color distortion issue. Fig. 6(e) and (h) shows the improved transmission map evaluated using CNN [33] and IASM [50], respectively. However, these maps are not so effective for sky regions. Fig. 6(i) shows the refined transmission map evaluated using the proposed model. The obtained transmission map achieves significant depth information than the others.

Fig. 7 shows the impact of mask size on estimated transmission maps and their respective haze-free images. Three

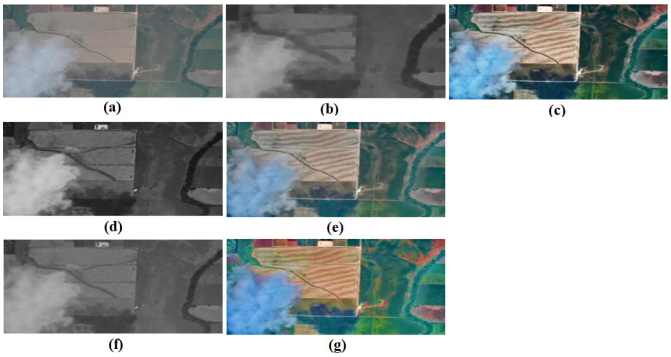


Fig. 7. Mask size analysis: (a) hazy image; (b) obtained transmission map and (c) obtained haze-free image by using $1 \times 1$ mask; (d) obtained transmission map and (e) obtained haze-free image by using $3 \times 3$ mask; and (f) obtained transmission map and (g) obtained haze-free image with $9 \times 9$ mask size.

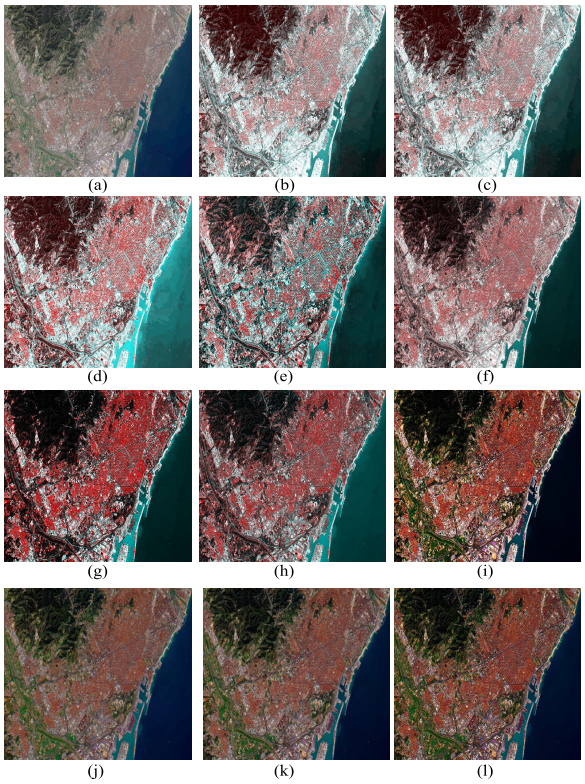


Fig. 8. Results of haze restoration models: (a) hazy image, (b) DCP [7], (c) PFAN [31], (d) AMEF [32], (e) CNN [33], (f) uDCP [39], (g) DFM [48], (h) IASM [50], (i) FDM [26], (j) PMS-Net [47], and (k) PHLD [49], and (l) Proposed.

different mask sizes are $1 \times 1$, $3 \times 3$, and $9 \times 9$ that are considered for experimental purpose. Fig. 7(b) and (c) shows that the size of mask is smaller, and then, the proposed haze removal model becomes less efficient. It is observed from Fig. 7(d) and (e) that the moderate mask size, i.e., $5 \times 5$, becomes more significant as the restored image has minimum saturated pixels, texture distortion, and halo and gradient reversal artifacts. Fig. 7(f) and (g) shows the results for larger mask size, i.e., $9 \times 9$. It is shown that the obtained restored image using $9 \times 9$ mask contains halo and gradient reversal artifacts.

Fig. 8 shows the visual comparative analysis of the proposed model with the existing haze removal models. It is

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 22:56:45 UTC from IEEE Xplore. Restrictions apply.

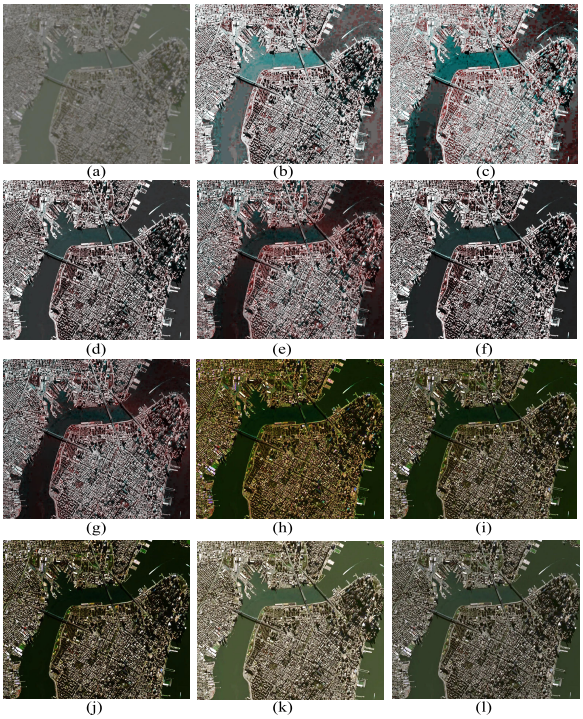


Fig. 9. Results of haze restoration models: (a) hazy image, (b) DCP [7], (c) PFAN [31], (d) AMEF [32], (e) CNN [33], (f) uDCP [39], (g) DFM [48], (h) IASM [50], (i) FDM [26], (j) PMS-Net [47], and (k) PHLD [49], and (l) Proposed.

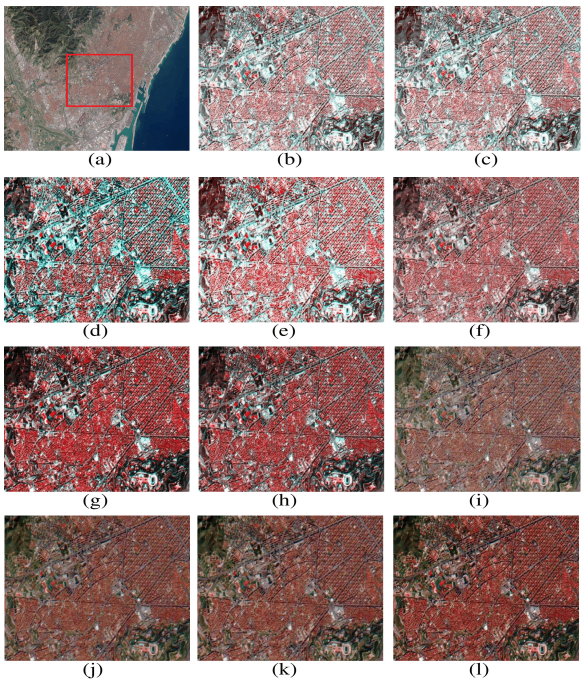


Fig. 10. Visual analysis of dehazing models on cropped image obtained from Fig. 8: (a) hazy image, (b) DCP [7], (c) PFAN [31], (d) AMEF [32], (e) CNN [33], (f) uDCP [39], (g) DFM [48], (h) IASM [50], (i) FDM [26], (j) PMS-Net [47], and (k) PHLD [49], and (l) Proposed.

found that DCP [7] and AMEF [32] suffer from texture distortion and edge degradation problems. The red marked squares show that the existing models still suffer from texture distortion problems. The obtained restored images from PFAN [31] and CNN [33] have better visibility compared to DCP [7] and AMEF [32]. However, these models suffer from edge and texture distortion problems. The restored images obtained from uDCP [39], DFM [48], IASM [50], and FDM [26] provide efficient results than DCP [7], AMEF [32], PFAN [31], and CNN [33]. However, these images show texture distortion and saturated pixels problems. PMS-Net [47] and PHLD [49] achieve remarkable results than DCP [7], PFAN [31], AMEF [32], CNN [33], uDCP [39], DFM [48], IASM [50], and FDM [26], but still some texture distortion is there. Compared to the existing models, the proposed model achieves better results by preserving texture, color, and edges of the obtained image.

Fig. 9(b) shows the restored image obtained from DCP [7], which suffers from saturated pixels, color distortion, edge distortion, spatial and spectral information, and gradient distortion distortion problems. In addition, the restored image seems to be overenhanced and different kinds of artifacts. The restored image obtained from PFAN [31] and CNN [33] is shown in Fig. 9(c) and (e). PFAN [31] and CNN [33] have achieved better visibility than DCP [7] and, however, still suffer from spatial, spectral, and texture distortions. Like DCP [7], PFAN [31] and CNN [33] suffer from the saturation of pixels. The haze-free images obtained from AMEF [32] and uDCP [39] achieve better results than DCP [7], PFAN [31], and CNN [33]. However, the major issues are color distortion, overenhancement, and texture distortion. The obtained restored images from DFM [48] and IASM [50] achieve better restored image than DCP [7], AMEF [32], PFAN [31], CNN [33], and uDCP [39]. However, these images still show oversaturation and texture distortion problems. PMS-Net [47] and FDM [26] achieve remarkable results than DCP [7], PFAN [31], AMEF [32], CNN [33], uDCP [39], DFM [48], and IASM [50]. It is shown that the restored images are more darker and also some texture distortion is also there. The results obtained from PHLD [49] [see Fig. 9(k)] and the proposed model [see Fig. 9(l)] have shown better texture, spatial, and spectral information than the existing models. However, compared to PHLD [49], the proposed model shows natural appearance with better spatial and spectral information.

Figs. 10 and 11 show the artifacts and distortion problems with the existing models by zooming two different small regions of Figs. 8 and 9, respectively. The zoomed area in the actual images is marked with red squares in Figs. 8(a) and 9(a). These figures prove that the images obtained from the competitive models contain some haze, spatial and spectral distortion, gradient reversal artifacts, saturation, and texture distortion kinds of problems. Figs. 10(l) and 11(l) shows the zoomed restored images obtained from the proposed model. It is found that the proposed model achieves remarkable visual performance with better texture information with minimum spectral and spatial distortion.

### *D. Quantitative Analysis*

This section presents the comparative analysis among the existing and the proposed haze removal models.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 22:56:45 UTC from IEEE Xplore. Restrictions apply.

TABLE III

QUANTITATIVE ANALYSIS IN TERMS OF PERCENTAGE OF SATURATED PIXELS ($\rho$), NEW VISIBLE EDGES ($e$), AND RATIO OF MEAN GRADIENT ($\bar{r}$)

| Dataset | Metric | DCP | PFAN | AMEF | CNN | uDCP | DFM | IASM | FDM | PMS-Net | PHLD | Prop. |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| DHazy | $\rho$ | 0.174 | 0.163 | 0.129 | 0.088 | 0.079 | 0.063 | 0.070 | 0.073 | 0.048 | 0.037 | **0.029** |
| | $e$ | 1.734 | 1.992 | 2.103 | 2.691 | 2.940 | 3.298 | 3.867 | 3.184 | 3.454 | **3.892** | 3.692 |
| | $\bar{r}$ | 1.644 | 1.762 | 1.907 | 2.103 | 2.203 | 2.396 | **2.602** | 2.183 | 2.348 | 2.463 | 2.542 |
| FRIDA | $\rho$ | 0.146 | 0.134 | 0.104 | 0.067 | 0.064 | 0.049 | 0.036 | 0.032 | 0.028 | 0.017 | **0.013** |
| | $e$ | 1.197 | 1.276 | 1.674 | 1.693 | 2.392 | 2.472 | 3.688 | 2.157 | 2.457 | 2.495 | **2.871** |
| | $\bar{r}$ | 0.944 | 1.204 | 1.283 | 1.461 | 1.681 | 1.928 | 2.194 | 2.057 | 2.158 | 2.243 | **2.282** |
| IVC | $\rho$ | 0.113 | 0.098 | 0.102 | 0.077 | 0.067 | 0.044 | 0.062 | 0.712 | 0.642 | **0.410** | 0.060 |
| | $e$ | 2.366 | 2.616 | 2.284 | 2.181 | 2.460 | 2.369 | 2.629 | 2.354 | 2.394 | 2.597 | **2.786** |
| | $\bar{r}$ | 1.233 | 1.616 | 1.904 | 2.111 | 2.188 | 2.216 | 2.023 | 1.982 | 2.107 | 2.194 | **2.217** |
| FRIDA2 | $\rho$ | 0.169 | 0.166 | 0.104 | 0.083 | **0.021** | 0.069 | 0.026 | 0.088 | 0.49 | 0.041 | **0.021** |
| | $e$ | 2.097 | 2.199 | 2.417 | 2.604 | 2.512 | 2.647 | 2.938 | 2.481 | 2.497 | 2.762 | **2.967** |
| | $\bar{r}$ | 2.084 | 2.161 | 2.483 | 2.682 | 2.788 | 2.748 | 2.446 | 2.258 | 2.425 | 2.459 | **2.806** |
| HSTS | $\rho$ | 0.179 | 0.137 | 0.122 | 0.081 | 0.070 | 0.042 | 0.029 | 0.074 | 0.060 | 0.034 | **0.011** |
| | $e$ | 2.330 | 2.499 | 2.389 | 2.228 | 2.197 | 2.244 | 2.592 | 2.248 | 2.514 | 2.645 | **2.674** |
| | $\bar{r}$ | 1.126 | 1.186 | 1.389 | 1.666 | 1.822 | 1.723 | 1.946 | 1.782 | 1.815 | **1.952** | 1.881 |
| RITS | $\rho$ | 0.160 | 0.144 | 0.161 | 0.098 | 0.080 | 0.071 | 0.061 | 0.057 | 0.038 | 0.036 | **0.029** |
| | $e$ | 1.641 | 1.680 | 1.692 | 1.489 | 2.167 | 2.204 | 2.263 | 2.183 | 2.217 | 2.241 | **2.360** |
| | $\bar{r}$ | 1.109 | 1.314 | 1.421 | 1.962 | 2.140 | 2.262 | 2.218 | 2.102 | 2.097 | 2.182 | **2.326** |
| SOTS | $\rho$ | 0.161 | 0.160 | 0.172 | 0.093 | 0.084 | 0.067 | 0.036 | 0.054 | 0.049 | 0.032 | **0.027** |
| | $e$ | 2.144 | 2.631 | 2.469 | 2.279 | 2.269 | 2.463 | 2.621 | 2.258 | 2.589 | .691 | **2.836** |
| | $\bar{r}$ | 1.864 | 1.783 | 2.216 | 2.331 | **2.402** | 2.098 | 1.928 | 2.021 | 1.975 | 2.082 | 2.143 |
| OTS | $\rho$ | 0.133 | 0.062 | 0.066 | 0.049 | 0.068 | 0.046 | 0.029 | 0.041 | 0.029 | 0.024 | **0.013** |
| | $e$ | 1.223 | 1.394 | 2.320 | 1.467 | 1.704 | 2.109 | 2.429 | 2.215 | 2.562 | 2.261 | **2.486** |
| | $\bar{r}$ | 1.306 | 1.708 | 1.906 | 2.162 | 2.143 | 2.106 | 2.282 | 2.204 | 2.218 | 2.268 | **2.297** |
| ITS | $\rho$ | 0.166 | 0.043 | 0.064 | 0.087 | 0.063 | 0.062 | 0.039 | 0.049 | 0.045 | 0.034 | **0.018** |
| | $e$ | 1.122 | 1.924 | 2.126 | 2.862 | 2.621 | 2.462 | 2.449 | 2.215 | 2.234 | 2.252 | **2.637** |
| | $\bar{r}$ | 0.962 | 1.303 | 1.204 | 1.634 | 1.792 | 2.081 | 1.773 | 1.724 | 1.878 | 1.892 | **1.936** |
| RW | $\rho$ | 0.181 | 0.114 | 0.144 | 0.098 | 0.046 | 0.042 | 0.063 | 0.059 | 0.052 | 0.041 | **0.036** |
| | $e$ | 1.684 | 1.923 | 2.381 | 2.139 | 2.678 | 2.396 | 2.314 | 2.225 | 2.245 | 2.268 | **2.732** |
| | $\bar{r}$ | 1.162 | 1.204 | 1.691 | 1.998 | 1.803 | 2.116 | 2.106 | 2.032 | 2.112 | **2.231** | 2.161 |

TABLE IV

QUANTITATIVE ANALYSIS IN TERMS OF PEAK SIGNAL-TO-NOISE RATIO, SSIM, AND ET

| Dataset | Metric | DCP | PFAN | AMEF | CNN | uDCP | DFM | IASM | FDM | PMS-Net | PHLD | Prop. |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| DHazy | PSNR | 17.30 | 17.72 | 18.26 | 18.77 | 19.09 | 19.66 | 20.41 | 19.15 | 20.23 | 20.97 | **21.63** |
| | SSIM | 0.781 | 0.812 | 0.823 | 0.836 | 0.861 | 0.868 | 0.863 | 0.868 | 0.876 | 0.879 | **0.884** |
| | ET | 89.14 | 6.74 | 8.32 | 6.29 | 7.86 | 6.41 | 1.69 | 3.48 | 2.12 | 1.26 | **1.13** |
| FRIDA | PSNR | 17.40 | 16.90 | 17.60 | 17.90 | 18.09 | 20.09 | 21.10 | 19.78 | 20.68 | 21.27 | **21.49** |
| | SSIM | 0.763 | 0.796 | 0.764 | 0.796 | 0.809 | 0.826 | 0.841 | 0.836 | 0.843 | 0.849 | **0.862** |
| | ET | 98.66 | 9.13 | 9.86 | 7.36 | 7.21 | 6.18 | 1.98 | 4.28 | 1.98 | 1.78 | **1.66** |
| IVC | PSNR | 18.08 | 18.34 | 18.89 | 19.40 | 19.86 | 20.08 | 20.39 | 19.87 | 21.24 | 21.37 | **21.48** |
| | SSIM | 0.772 | 0.776 | 0.792 | 0.804 | 0.826 | 0.846 | 0.867 | 0.845 | 0.879 | 0.882 | **0.893** |
| | ET | 86.76 | 8.63 | 8.60 | 7.06 | 7.34 | 6.89 | 1.79 | 3.48 | 1.79 | 1.68 | **1.34** |
| FRIDA2 | PSNR | 18.20 | 18.40 | 17.96 | 19.11 | 18.49 | 20.72 | 20.92 | 20.65 | 21.58 | 21.87 | **22.16** |
| | SSIM | 0.806 | 0.824 | 0.843 | 0.828 | 0.834 | 0.864 | 0.869 | 0.842 | 0.858 | 0.862 | **0.873** |
| | ET | 84.76 | 8.63 | 8.18 | 7.62 | 7.36 | 6.91 | 1.66 | 4.08 | 1.91 | 1.82 | **1.62** |
| HSTS | PSNR | 16.20 | 16.60 | 17.09 | 17.63 | 19.34 | 20.89 | **21.16** | 19.64 | 19.95 | 20.87 | 21.08 |
| | SSIM | 0.762 | 0.766 | 0.749 | 0.796 | 0.839 | 0.849 | **0.872** | 0.828 | 0.846 | 0.851 | 0.863 |
| | ET | 86.84 | 8.63 | 8.61 | 4.61 | 7.37 | 6.91 | 1.76 | 3.58 | 2.05 | 1.78 | 1.23 |
| RITS | PSNR | 18.33 | 18.11 | 19.42 | 17.16 | 19.21 | 20.26 | 20.68 | 20.15 | 21.25 | 21.36 | **21.43** |
| | SSIM | 0.732 | 0.794 | 0.761 | 0.782 | 0.823 | 0.834 | 0.869 | 0.846 | 0.858 | 0.860 | **0.876** |
| | ET | 81.99 | 8.75 | 8.72 | 7.46 | 4.91 | 6.21 | 1.69 | 3.58 | 2.18 | 1.94 | **1.44** |
| SOTS | PSNR | 16.26 | 17.32 | 17.16 | 18.40 | **21.68** | 19.98 | 21.21 | 20.28 | 20.87 | 21.16 | 21.58 |
| | SSIM | 0.767 | 0.789 | 0.792 | 0.819 | 0.831 | **0.891** | 0.873 | 0.845 | 0.868 | 0.875 | 0.881 |
| | ET | 99.67 | 9.72 | 8.62 | 7.61 | 6.77 | 6.61 | 1.78 | 3.59 | 2.15 | 1.81 | **1.16** |
| OTS | PSNR | 17.16 | 18.26 | 19.33 | 19.66 | 20.11 | 20.23 | 21.14 | 20.15 | 21.62 | 21.88 | **22.73** |
| | SSIM | 0.796 | 0.816 | 0.837 | 0.847 | 0.864 | 0.874 | 0.831 | 0.835 | 0.862 | 0.864 | **0.878** |
| | ET | 86.96 | 9.62 | 8.72 | 7.19 | 7.69 | 6.15 | 1.69 | 3.28 | 1.98 | 1.92 | **1.73** |
| ITS | PSNR | 16.60 | 17.42 | 18.22 | 19.62 | 19.28 | 20.63 | 20.90 | 20.25 | 21.67 | 22.58 | **23.18** |
| | SSIM | 0.783 | 0.801 | 0.816 | 0.836 | 0.829 | 0.845 | 0.863 | 0.843 | 0.862 | 0.869 | **0.883** |
| | ET | 98.87 | 8.62 | 8.61 | 7.98 | 6.76 | 7.68 | 1.82 | 2.97 | 1.98 | 1.82 | **1.72** |
| RW | PSNR | 16.31 | 16.68 | 17.62 | 18.02 | 18.49 | 18.84 | 19.246 | 18.58 | 19.25 | 19.63 | **20.41** |
| | SSIM | 0.763 | 0.782 | 0.796 | 0.802 | 0.816 | 0.836 | 0.871 | 0.846 | 0.868 | 0.872 | **0.889** |
| | ET | 87.67 | 8.93 | 8.72 | 7.79 | 6.16 | 6.89 | 1.68 | 3.12 | 1.63 | 1.59 | **1.32** |

Table III shows the quantitative analysis of the proposed model in terms of percentage of saturated pixels ($\rho$), new visible edges ($e$), and ratio of mean gradient ($\bar{r}$). Bold values indicate higher performance. $\rho$ computes the pixels that become saturated after applying the haze restoration model, and therefore, it should be minimized. Table III

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 22:56:45 UTC from IEEE Xplore. Restrictions apply.

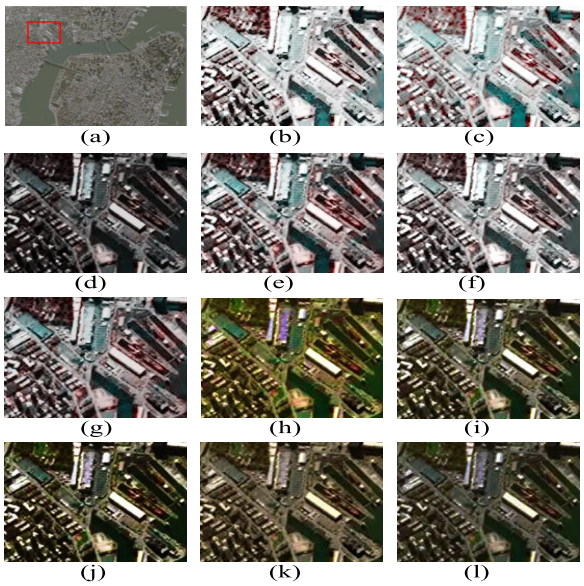


Fig. 11. Visual analysis of dehazing models on cropped image obtained from Fig. 9: (a) hazy image, (b) DCP [7], (c) PFAN [31], (d) AMEF [32], (e) CNN [33], (f) uDCP [39], (g) DFM [48], (h) IASM [50], (i) FDM [26], (j) PMS-Net [47], and (k) PHLD [49], and (l) Proposed.

demonstrates that the proposed haze removal model provides significantly lesser saturated values than the existing models. The computed mean reduction is ≈0.0276.

Hautiere *et al.* [60] proposed two performance metrics, i.e., new visible edges ($e$) and mean gradient ($\bar{r}$). $e$ shows that the visibility restoration model can preserve edges efficiently. $\bar{r}$ shows the gradients of the obtained image. It states that the obtained image has a good gradient and texture information. From Table III, it is observed that the proposed haze removal model attains good restoration results than the other models as it achieves maximum values of $e$ and $\bar{r}$. The computed mean improvements in terms of $e$ and $\bar{r}$ are ≈0.1342 and ≈0.1292, respectively.

Table IV shows the quantitative analysis of the proposed model in terms of peak signal-to-noise ratio (in dB), structural similarity index metric (SSIM), and execution time (ET). PSNR is computed between ground-truth and restore images. From Table IV, it is found that the mean enhancement in PSNR by using the proposed haze removal model than the existing haze restoration models is 1.1659. From Table IV, it is observed that the mean enhancement in SSIM by using the proposed haze removal model over the existing models is 0.0176. Table IV also shows the ET analysis (in seconds). The proposed haze removal model restores images at significantly good computational speed than the existing models. The mean reduction in the ET of the proposed haze removal model than the existing models is ≈0.2498.

Table V shows the ET analysis of the proposed model by varying the image size. tic and toc operators of MATLAB are used to estimate the computation time of dehazing models. We have scaled the images to various dimensions, such as $256 \times 256$, $512 \times 512$, $1024 \times 1024$, $2048 \times 2048$, $4096 \times 4096$, and $8192 \times 8192$. Median $\pm$ IQR $\times$ 1.5 values are computed. It is found that the proposed model can restore even high size remote sensing images with good computational speed.

TABLE V
ET ANALYSIS OF THE PROPOSED MODEL BY VARYING THE IMAGE SIZE

| Image size | Time (in seconds) |
|---|---|
| $256 \times 256$ | $0.87 \pm 0.31$ |
| $512 \times 512$ | $1.43 \pm 0.54$ |
| $1024 \times 1024$ | $3.64 \pm 1.19$ |
| $2048 \times 2048$ | $11.56 \pm 2.82$ |
| $4096 \times 4096$ | $24.65 \pm 4.48$ |
| $8192 \times 8192$ | $52.48 \pm 10.34$ |

## VI. CONCLUSION

An evolving visibility restoration model was proposed to restore the haze from remote sensing images. Initially, an efficient fusion of transmission maps was proposed by using OGPP and luminance-based channels prior. The fused transmission map was further improved by using a hybrid constraint-based variational model. Finally, dynamic differential evolution was utilized to evolve the control parameters of the proposed model. The proposed and ten existing image dehazing models were tested and validated on benchmark synthetic and real-life hazy images. The comparative visual analysis demonstrated that the proposed visibility restoration model preserves color, edge, texture information, and spectral and spatial information more efficiently than the existing models. Quantitative analysis revealed that the proposed model can be used as a preprocessing model for real-time remote sensing imaging applications.

In the near future, the proposed work will be extended by designing a novel optimization model for the efficient selection of control parameters. In addition, an improved oblique gradient channel prior can be designed by increasing the mask size.

## REFERENCES

[1] Z. Gu, Z. Zhan, Q. Yuan, and L. Yan, "Single remote sensing image dehazing using a prior-based dense attentive network," *Remote Sens.*, vol. 11, no. 24, p. 3008, Dec. 2019.
[2] C. O. Ancuti, C. Ancuti, and P. Bekaert, "Enhancing by saliency-guided decolorization," in *Proc. CVPR*, Jun. 2011, pp. 257–264.
[3] M. Jian, Y. Yin, J. Dong, and W. Zhang, "Comprehensive assessment of non-uniform illumination for 3D heightmap reconstruction in outdoor environments," *Comput. Ind.*, vol. 99, pp. 110–118, Aug. 2018.
[4] K. B. Gibson, "An investigation of dehazing effects on image and video coding," *IEEE Trans. Image Process.*, vol. 21, no. 2, pp. 662–673, Jan. 2012.
[5] T. Gao *et al.*, "A novel UAV sensing image defogging method," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 13, pp. 2610–2625, 2020.
[6] X. Li, H. Shen, L. Zhang, H. Zhang, Q. Yuan, and G. Yang, "Recovering quantitative remote sensing products contaminated by thick clouds and shadows using multitemporal dictionary learning," *IEEE Trans. Geosci. Remote Sens.*, vol. 52, no. 11, pp. 7086–7098, Nov. 2014.
[7] K. He, J. Sun, and X. Tang, "Single image haze removal using dark channel prior," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 33, no. 12, pp. 2341–2353, Sep. 2011.
[8] A. Alajarmeh and A. A. Zaidan, "A real-time framework for video dehazing using bounded transmission and controlled Gaussian filter," *Multimedia Tools Appl.*, vol. 77, no. 20, pp. 26315–26350, Oct. 2018.
[9] S. Liu and Y. Zhang, "Detail-preserving underexposed image enhancement via optimal weighted multi-exposure fusion," *IEEE Trans. Consum. Electron.*, vol. 65, no. 3, pp. 303–311, Aug. 2019.
[10] Q. Zhu, J. Mai, and L. Shao, "A fast single image haze removal algorithm using color attenuation prior," *IEEE Trans. Image Process.*, vol. 24, no. 11, pp. 3522–3533, Nov. 2015.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 22:56:45 UTC from IEEE Xplore. Restrictions apply.

[11] X. Liu, H. Zhang, Y. Y. Tang, and J. X. Du, "Scene-adaptive single image dehazing via opening dark channel model," *IET Image Process.*, vol. 10, no. 11, pp. 877–884, 2016.

[12] I. Riaz, X. Fan, and H. Shin, "Single image dehazing with bright object handling," *IET Comput. Vis.*, vol. 10, no. 8, pp. 817–827, Dec. 2016.

[13] S. G. Narasimhan and S. K. Nayar, "Contrast restoration of weather degraded images," *IEEE Trans. Pattern Anal. Mach. Learn.*, vol. 25, no. 6, pp. 713–724, Jun. 2003.

[14] S. K. Nayar and S. G. Narasimhan, "Vision in bad weather," in *Proc. 7th IEEE Int. Conf. Comput. Vis.*, vol. 2, Sep. 1999, pp. 820–827.

[15] B. Li, S. Wang, J. Zheng, and L. Zheng, "Single image haze removal using content-adaptive dark channel and post enhancement," *IET Compt. Vis.*, vol. 8, no. 2, pp. 131–140, 2014.

[16] D. Wang and J. Zhu, "Fast smoothing technique with edge preservation for single image dehazing," *IET Comput. Vis.*, vol. 9, no. 6, pp. 950–959, 2015.

[17] J. P. Oakley and B. L. Satherley, "Improving image quality in poor visibility conditions using a physical model for contrast degradation," *IEEE Trans. Image Process.*, vol. 7, no. 2, pp. 167–179, Feb. 1998.

[18] D. Nair and P. Sankaran, "Color image dehazing using surround filter and dark channel prior," *J. Vis. Commun. Image Represent.*, vol. 50, pp. 9–15, Jan. 2018.

[19] J. Bala and K. Lakhwani, "Single image desmogging using oblique gradient profile prior and variational minimization," *Multidimensional Syst. Signal Process.*, vol. 31, pp. 1259–1275, Feb. 2020.

[20] A. Mathias and D. Samiappan, "Underwater image restoration based on diffraction bounded optimization algorithm with dark channel prior," *Optik*, vol. 192, Sep. 2019, Art. no. 162925.

[21] Y. Wang, T.-Z. Huang, X.-L. Zhao, L.-J. Deng, and T.-Y. Ji, "A convex single image dehazing model via sparse dark channel prior," *Appl. Math. Comput.*, vol. 375, Jun. 2020, Art. no. 125085.

[22] Y. Jiang, C. Sun, Y. Zhao, and L. Yang, "Image dehazing using adaptive bi-channel priors on superpixels," *Comput. Vis. Image Understand.*, vol. 165, pp. 17–32, Dec. 2017.

[23] M. Zhu, B. He, J. Liu, and J. Yu, "Boosting dark channel dehazing via weighted local constant assumption," *Signal Process.*, vol. 171, Jun. 2020, Art. no. 107453.

[24] M. Kaur, D. Singh, V. Kumar, and K. Sun, "Color image dehazing using gradient channel prior and guided L0 filter," *Inf. Sci.*, vol. 521, pp. 326–342, Jun. 2020.

[25] G. Hou, J. Li, G. Wang, H. Yang, B. Huang, and Z. Pan, "A novel dark channel prior guided variational framework for underwater image restoration," *J. Vis. Commun. Image Represent.*, vol. 66, Jan. 2020, Art. no. 102732.

[26] J. M. Guo, J.-Y. Syue, V. R. Radzicki, and H. Lee, "An efficient fusion-based defogging," *IEEE Trans. Image Process.*, vol. 26, no. 9, pp. 4217–4228, Sep. 2017.

[27] S. M. Yoon, "Visibility enhancement of fog-degraded image using adaptive total variation minimisation," *Imag. Sci. J.*, vol. 64, no. 2, pp. 82–86, Feb. 2016.

[28] C. O. Ancuti and C. Ancuti, "Single image dehazing by multi-scale fusion," *IEEE Trans. Image Process.*, vol. 22, no. 8, pp. 3271–3282, Aug. 2013.

[29] W. Wang, X. Yuan, X. Wu, and Y. Liu, "Dehazing for images with large sky region," *Neurocomputing*, vol. 238, pp. 365–376, May 2017.

[30] W. Wang, X. Yuan, X. Wu, and Y. Liu, "Fast image dehazing method based on linear transformation," *IEEE Trans. Multimedia*, vol. 19, no. 6, pp. 1142–1155, Jun. 2017.

[31] X. Zhang, T. Wang, J. Wang, G. Tang, and L. Zhao, "Pyramid channel-based feature attention network for image dehazing," *Comput. Vis. Image Understand.*, vols. 197–198, Aug. 2020, Art. no. 103003.

[32] M. Zheng, G. Qi, Z. Zhu, Y. Li, H. Wei, and Y. Liu, "Image dehazing by an artificial image fusion method based on adaptive structure decomposition," *IEEE Sensors J.*, vol. 20, no. 14, pp. 8062–8072, Jul. 2020.

[33] C. Li, C. Guo, J. Guo, P. Han, H. Fu, and R. Cong, "PDR-net: Perception-inspired single image dehazing network with refinement," *IEEE Trans. Multimedia*, vol. 22, no. 3, pp. 704–716, Mar. 2020.

[34] S. Zhang, F. He, W. Ren, and J. Yao, "Joint learning of image detail and transmission map for single image dehazing," *Vis. Comput.*, vol. 36, no. 2, pp. 305–316, Feb. 2020.

[35] M. Ju, C. Ding, Y. J. Guo, and D. Zhang, "IDGCP: Image dehazing based on gamma correction prior," *IEEE Trans. Image Process.*, vol. 29, pp. 3104–3118, 2020.

[36] D. Zhao, L. Xu, Y. Yan, J. Chen, and L.-Y. Duan, "Multi-scale Optimal Fusion model for single image dehazing," *Signal Process., Image Commun.*, vol. 74, pp. 253–265, May 2019.

[37] S. E. Kim, T. H. Park, and I. K. Eom, "Fast single image dehazing using saturation based transmission map estimation," *IEEE Trans. Image Process.*, vol. 29, pp. 1985–1998, 2020.

[38] P. Huang, L. Zhao, R. Jiang, T. Wang, and X. Zhang, "Self-filtering image dehazing with self-supporting module," *Neurocomputing*, vol. 432, pp. 57–69, Apr. 2021.

[39] A. Golts, D. Freedman, and M. Elad, "Unsupervised single image dehazing using dark channel prior loss," *IEEE Trans. Image Process.*, vol. 29, pp. 2692–2701, 2019.

[40] C. O. Ancuti, C. Ancuti, C. De Vleeschouwer, and M. Sbetr, "Color channel transfer for image dehazing," *IEEE Signal Process. Lett.*, vol. 26, no. 9, pp. 1413–1417, Sep. 2019.

[41] S. Yin, Y. Wang, and Y.-H. Yang, "A novel image-dehazing network with a parallel attention block," *Pattern Recognit.*, vol. 102, Jun. 2020, Art. no. 107255.

[42] T. Wang, L. Zhao, P. Huang, X. Zhang, and J. Xu, "Haze concentration adaptive network for image dehazing," *Neurocomputing*, vol. 439, pp. 75–85, Jun. 2021.

[43] Y. Shuai, R. Liu, and W. He, "Image haze removal of Wiener filtering based on dark channel prior," in *Proc. 8th Int. Conf. Comput. Intell. Secur.*, Nov. 2012, pp. 318–322.

[44] T. H. Kil, S. H. Lee, and N. I. Cho, "A dehazing algorithm using dark channel prior and contrast enhancement," in *Proc. IEEE Int. Conf. Acoust., Speech Signal Process.*, May 2013, pp. 2484–2487.

[45] T. M. Bui, H. N. Tran, W. Kim, and S. Kim, "Segmenting dark channel prior in single image dehazing," *Electron. Lett.*, vol. 50, no. 7, pp. 516–518, Mar. 2014.

[46] J.-B. Wang, N. He, L.-L. Zhang, and K. Lu, "Single image dehazing with a physical model and dark channel prior," *Neurocomputing*, vol. 149, pp. 718–728, Feb. 2015.

[47] W.-T. Chen, J.-J. Ding, and S.-Y. Kuo, "PMS-net: Robust haze removal based on patch map for single images," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2019, pp. 11673–11681.

[48] Y. Gao, Q. Li, and J. Li, "Single image dehazing via a dual-fusion method," *Image Vis. Comput.*, vol. 94, Feb. 2020, Art. no. 103868.

[49] W.-T. Chen, H.-Y. Fang, J.-J. Ding, and S.-Y. Kuo, "PMHLD: Patch map-based hybrid learning DehazeNet for single image haze removal," *IEEE Trans. Image Process.*, vol. 29, pp. 6773–6788, 2020.

[50] M. Ju, C. Ding, W. Ren, Y. Yang, D. Zhang, and Y. J. Guo, "IDE: Image dehazing and exposure using an enhanced atmospheric scattering model," *IEEE Trans. Image Process.*, vol. 30, pp. 2180–2192, 2021.

[51] Y. Gao, Y. Su, Q. Li, H. Li, and J. Li, "Single image dehazing via self-constructing image fusion," *Signal Process.*, vol. 167, Feb. 2020, Art. no. 107284.

[52] F. Guo, X. Zhao, J. Tang, H. Peng, L. Liu, and B. Zou, "Single image dehazing based on fusion strategy," *Neurocomputing*, vol. 378, pp. 9–23, Feb. 2020.

[53] D. Singh and V. Kumar, "A novel dehazing model for remote sensing images," *Comput. Electr. Eng.*, vol. 69, pp. 14–27, Jul. 2018.

[54] D. Singh and V. Kumar, "Image dehazing using Moore neighborhood-based gradient profile prior," *Signal Process., Image Commun.*, vol. 70, pp. 131–144, Feb. 2019.

[55] H. Lu, Q. Liu, M. Zhang, Y. Wang, and X. Deng, "Gradient-based low rank method and its application in image inpainting," *Multimedia Tools Appl.*, vol. 77, no. 5, pp. 5969–5993, Mar. 2018.

[56] Q. Shu, C. Wu, Z. Xiao, and R. W. Liu, "Variational regularized transmission refinement for image dehazing," in *Proc. IEEE Int. Conf. Image Process. (ICIP)*, Sep. 2019, pp. 2781–2785.

[57] B. I. Yun and M. S. Petković, "Iterative methods based on the signum function approach for solving nonlinear equations," *Numer. Algorithms*, vol. 52, no. 4, pp. 649–662, Dec. 2009.

[58] J. Bala and K. Lakhwani, "Performance evaluation of various desmogging techniques for single smoggy images," *Mod. Phys. Lett. B*, vol. 33, no. 5, Feb. 2019, Art. no. 1950056.

[59] A. K. Tripathi and S. Mukhopadhyay, "Removal of fog from images: A review," *IETE Tech. Rev.*, vol. 29, no. 2, pp. 148–156, Apr. 2012.

[60] N. Hautière, J.-P. Tarel, D. Aubert, and É. Dumont, "Blind contrast enhancement assessment by gradient ratioing at visible edges," *Image Anal. Stereol. J.*, vol. 27, no. 2, pp. 87–95, Jun. 2008.

[61] R. Storn and K. Price, "Differential evolution–A simple and efficient heuristic for global optimization over continuous spaces," *J. Global Optim.*, vol. 11, no. 4, pp. 341–359, 1997.

[62] Q. Fan and X. Yan, "Multi-objective modified differential evolution algorithm with archive-base mutation for solving multi-objective *p*-xylene oxidation process," *J. Intell. Manuf.*, vol. 29, no. 1, pp. 35–49, 2018.


Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 22:56:45 UTC from IEEE Xplore. Restrictions apply.

[63] Z.-K. Li, J.-R. Tan, Y.-X. Feng, and H. Fang, "Multi-objective particle swarm optimization algorithm based on crowding distance sorting and its application," *Comput. Integr. Manuf. Syst.*, vol. 7, pp. 1329–1336, Jan. 2008.
[64] (SIC). (2020). *Satellite Sensors*. [Online]. Available: https://www.satimagingcorp.com/satellite-sensors/
[65] B. Li *et al.*, "Benchmarking single image dehazing and beyond," 2017, *arXiv:1712.04143*.

**Dilbag Singh** (Senior Member, IEEE) received the Ph.D. degree in computer science and engineering from the Thapar Institute of Engineering and Technology, Patiala, India, in 2019.

He has worked as an Assistant Professor in three well-known universities of India, i.e., Chandigarh University, Kochi, India, Manipal University Jaipur, Jaipur, India, and Bennett University, Greater Noida, India. He has published more than 70 research articles in SCI/SCIE indexed journals. He has also submitted five patents and has published three books and two book chapters. His H-index is 29. His research areas include image processing, computer vision, deep learning, metaheuristic techniques, and information security.

Dr. Singh has acted as a Lead Guest Editor/an Editorial Board Member of many SCI/SCIE indexed journals, such as *Journal of Healthcare Engineering*, *Mathematical Problems in Engineering*, and *Journal of Intelligent & Fuzzy Systems*. He was in the top 2% list issues by "World Ranking of Top 2% Scientists" in 2021. He was part of the 11 Web of Science/Scopus indexed conferences.

**Manjit Kaur** (Member, IEEE) received the Master of Engineering degree in information technology from Punjab University, Chandigarh, India, in 2011, and the Ph.D. degree from the Thapar Institute of Engineering and Technology, Patiala, India, in 2019.

She is currently working as an Assistant Professor with the School of Engineering and Applied Sciences, Bennett University, Greater Noida, India. She has published more than 60 SCI/SCIE indexed articles so far. Her research interests include wireless sensor networks, digital image processing, and metaheuristic techniques.

Dr. Kaur was in the top 2% list issues by "World Ranking of Top 2% Scientists" in 2021. She was part of the 11 Web of Science/Scopus indexed conferences.

**Mohamed Yaseen Jabarulla** (Member, IEEE) received the B.E. degree in electronics and communications engineering from Anna University, Chennai, India, in 2011, and the M.S. degree in information and communications and the Ph.D. degree in electrical engineering and computer science from the Gwangju Institute of Science and Technology (GIST), Gwangju, South Korea, in 2016 and 2020, respectively.

He is currently working as a Post-Doctoral Researcher at the School of Electrical Engineering and Computer Science, GIST He is the author or coauthor of more than ten SCI/SCIE indexed journals. His research interests include medical image computing, computational imaging, artificial intelligence, blockchain, eHealth, and biomedical signal processing.

Dr. Jabarulla is a reviewer of more than 15 well-reputed journals.

**Vijay Kumar** received the B.Tech. degree from the Guru Jambheshwar University of Science and Technology, Hisar, India, in 2005, the M.Tech. degree from the M. M. Engineering College, Mullana, India, in 2008, and the Ph.D. degree from NIT Kurukshetra, Haryana, India, in 2015.

He is currently an Assistant Professor with the Computer Science and Engineering Department, NIT Hamirpur, Hamirpur, India. He has four years of teaching and research experience at the Thapar Institute of Engineering and Technology, Patiala, India. He has eight years of teaching experience in various reputed institutes. He completed two DST SERB-sponsored research projects. Presently, he is working on one CSIR sponsored research project. He has published more than 100 research papers in international journals/conferences. He has many book chapters in international repute publishers. He has supervised many Ph.D. and M.Tech. theses on metaheuristics, image mining, and data clustering. His research areas are soft computing, data mining, deep learning, steganography, and pattern recognition.

Dr. Kumar is a reviewer of several reputed SCI journals. He is a member of ACM, CSI, the International Association of Engineers, and the International Association of Computer Science and Information Technology, Singapore.

**Heung-No Lee** (Senior Member, IEEE) received the B.S., M.S., and Ph.D. degrees in electrical engineering from the University of California at Los Angeles, Los Angeles, CA, USA, in 1993, 1994, and 1999, respectively.

He was with HRL Laboratories, LLC, Malibu, CA, USA, as a Research Staff Member from 1999 to 2002. From 2002 to 2008, he was an Assistant Professor with the University of Pittsburgh, Pittsburgh, PA, USA. In 2009, he moved to the School of Electrical Engineering and Computer Science, Gwangju Institute of Science and Technology, Gwangju, South Korea. He is currently with the School of Electrical Engineering and Computer Science, Gwangju Institute of Science and Technology. His research interests include information theory, signal processing theory, blockchain, communications/networking theory, and their application to wireless communications and networking, compressive sensing, future Internet, and brain-computer interface.

Dr. Lee received several prestigious national awards, including the Top 100 National Research and Development Award in 2012, the Top 50 Achievements of Fundamental Researches Award in 2013, and the Science/Engineer of the Month in January 2014.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 22:56:45 UTC from IEEE Xplore. Restrictions apply.